

2019 Ph H2 Q5

Section: Particles and Waves

Topic: Doppler Effect — Car sound and medical ultrasound

(a) A car moving at 12 m s^{-1} emits sound of frequency 510 Hz. An observer at the roadside hears the frequency change as the car passes. (i) Name the effect. (ii) Calculate the frequency heard as the car approaches. (b) Ultrasound Doppler test for red blood cell speed. Probe emits ultrasound at 3.70 MHz, velocity 1540 m s^{-1} , angle 60° , $\Delta f = 286 \text{ Hz}$. Calculate velocity of red blood cells.

Worked solution

(a)(i) The effect is called the Doppler effect — the observed frequency of a wave changes if there is relative motion between source and observer.

(a)(ii) For a moving source and stationary observer:

$$\begin{aligned} f' &= f (v / (v - v_s)). \\ &= 510 \times (340 / (340 - 12)). \\ &= 528.7 \text{ Hz}. \end{aligned}$$

Answer: 528 Hz

(b) Doppler ultrasound formula: $\Delta f = (2 f v_{\text{rbc}} \cos\theta)/v$.

Rearrange: $v_{\text{rbc}} = \Delta f v / (2 f \cos\theta)$.

$$\begin{aligned} &= 286 \times 1540 / (2 \times 3.70 \times 10^6 \times \cos 60^\circ). \\ &= 0.12 \text{ m s}^{-1}. \end{aligned}$$

Answer: 0.06 m s^{-1}

Final answers

(a)(i) Doppler effect

(a)(ii) $f' \approx 528 \text{ Hz}$

(b) $v_{\text{rbc}} \approx 0.06 \text{ m s}^{-1}$

Revision tips

- Doppler effect: apparent frequency increases when source approaches.
- Formula for moving source: $f' = f(v/(v-v_s))$.
- Ultrasound Doppler uses frequency shift from moving blood cells.
- $\Delta f = 2f v_{\text{rbc}} \cos\theta / v$.
- Small Δf means blood flow speed is small compared with ultrasound speed.